

Unified Batch & Real-Time Data Ingestion into Snowflake

The Challenge

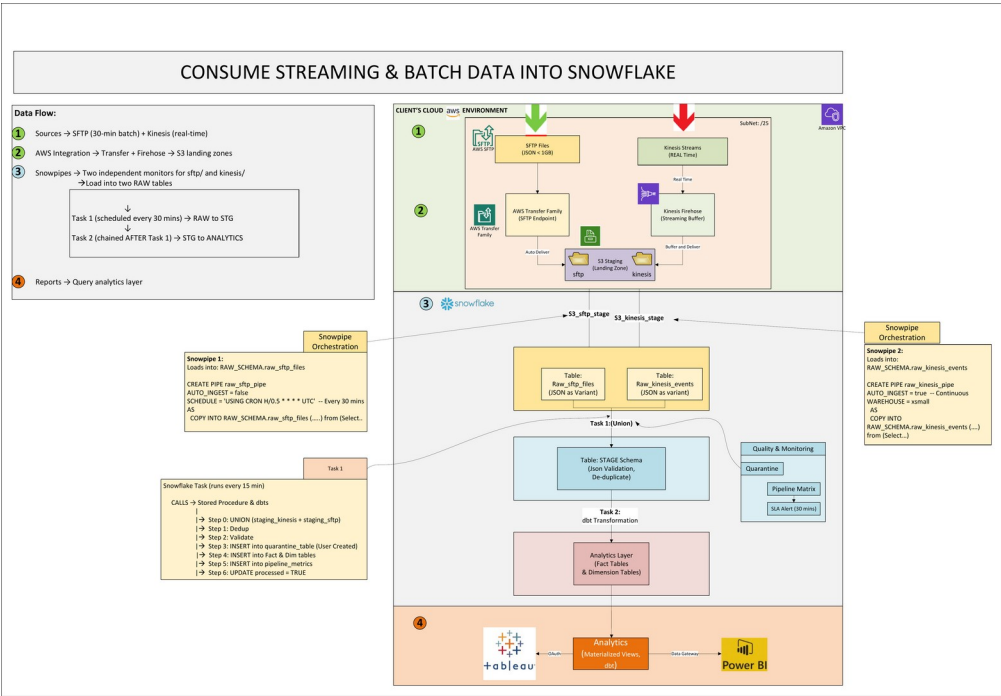
A client needed to consolidate SFTP batch files (delivered twice daily, <1GB compressed) and near real-time Kinesis event streams — both in the same JSON format — into production-ready Snowflake tables within a 30-minute SLA.

The Solution

- AWS Transfer Family + Kinesis Firehose land both sources into separate S3 zones
- Two independent Snowpipes load raw JSON (VARIANT) on their own cadence — 30-min schedule for SFTP, continuous auto-ingest for Kinesis
- Chained Snowflake Tasks union, dedupe, validate, and quarantine bad records every 15 minutes
- dbt transforms clean data into Fact/Dimension tables for Tableau & Power BI
- Snowflake PrivateLink + VPC endpoints keep all traffic off the public internet

Outcome

A single, modular pipeline unifying batch and streaming sources through shared transformation logic — comfortably inside the 30-minute SLA, with built-in quarantine, metrics, and SLA alerting.



Key Decision: Streaming Ingestion Method

Criteria	Firehose → S3 → Snowpipe (Selected)	Snowpipe Streaming
Latency	~5 min	~seconds
Reliability	High	Medium
Data Loss Risk	Near zero (S3 durable)	Lambda failures
Best When	SLA is 5–30 min, want simplicity	Need sub-minute latency

Considerations for production hardening: Firehose's ~5-min buffering consumes part of the SLA budget; dedup logic must cover retries on either path; schema drift needs explicit validation; ingestion cost should be monitored as volume scales.